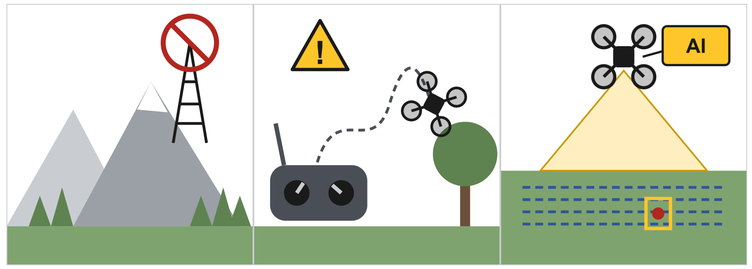
XX-000

KESTREL: Just Say Who's Missing

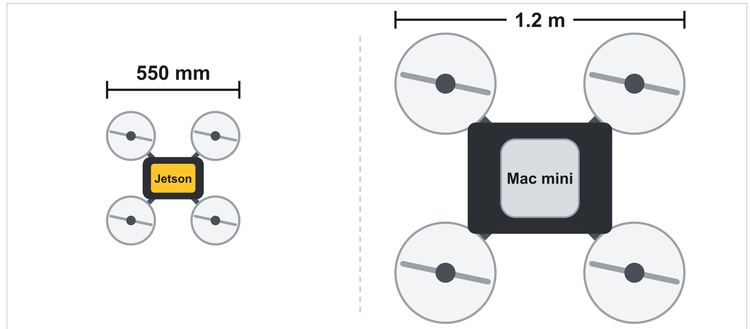
Shiva Shrestha and Honghui Xu

The Problem



No signal | Hard to fly | KESTREL flies itself

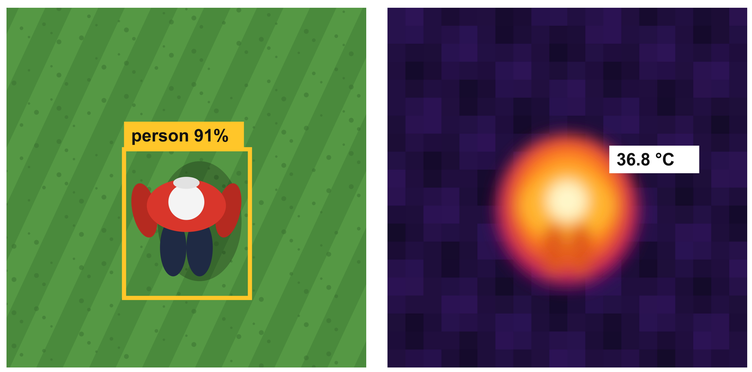
Two Drone Models



Compact, 550 mm
Jetson · lighter AI

Large, 1.2 m
Mac mini · bigger AI

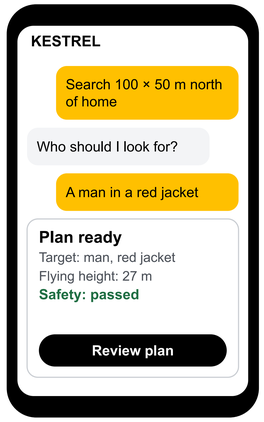
Finding the Person

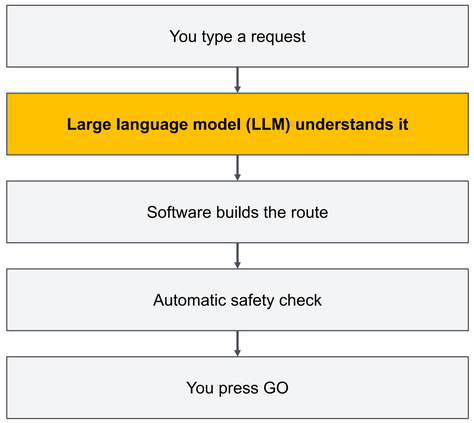


Colour camera | Thermal camera

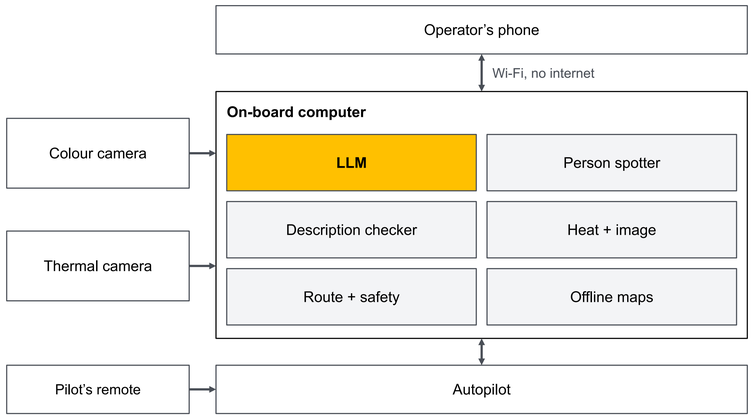
MATCH → photo + map pin to phone

Plan a Search by Chatting

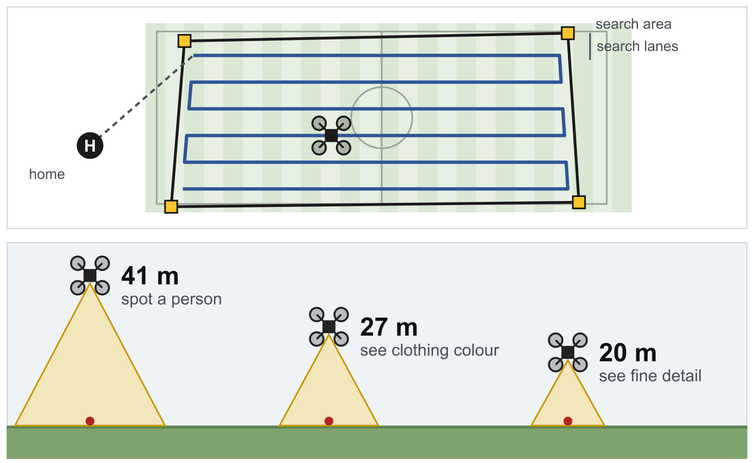




System Overview



Search Pattern and Height

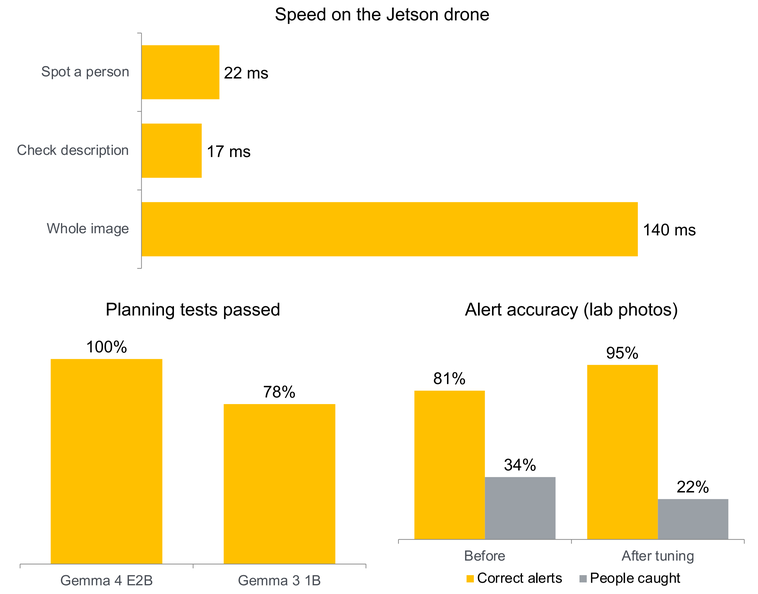


Automatic Safety Checks

- ✓ GPS locked
- ✓ Virtual fence
- ✓ Airspace approval
- ✓ Autopilot self-test
- ✓ Cameras and AI
- ✓ Battery level
- ✓ 400 ft limit
- ✓ Pilot's remote
- ✓ Route verified
- ✓ Legal checklist

FAA Part 107 · licensed pilot can take over

Results



631
software tests

0
internet needed

2
drone models

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